Computer Science & Information Technology

Computer Networks

DPP: 1

Flow Control

Q1 Consider two hosts are connected directly through point-to-point link. The link bandwidth is 1Mbps, distance is 10 km, signal propagation speed is 20 millisecond per km and packet size is 2000 Bytes than calculate transmission and propagation delay (in millisecond)?

(A) 2 and 200 (B) 8 and 20
(C) 16 and 20 (D) 16 and 200

Q2 Consider two hosts S and D are connected through one ethernet switch, switch takes 15 ms to process a packet. Each link bandwidth is 1 Mbps, distance is 20 km, signal propagation speed is $2 * 10^5$ meter per sec and packet size is 8000 bytes than calculate how much time (in millisecond) it takes to transmits a file of size 24000 bytes from S to D?

Q3 Consider two hosts S and D are connected through two intermediate routers, router takes negligibal processing time. Each link bandwidth is 100 Kbps, distance is 10 km, signal propagation speed is 5 milisecond per km and packet size is 1000 bits than calculate how much time (in millisecond) it takes to transmits a file of size 4000 bytes from S to D?

Q4 Consider two hosts connected directly through point-to-point link uses Stop-and-Wait ARQ for flow control. The link bandwidth is 1Mbps and packet size is 1000 bytes. To achieve 50 % utilization, what should be propagation delay (in millisecond)?

(A) 2 (B) 4
(C) 6 (D) 8

Q5 Consider two hosts connected directly through point-to-point link uses Stop-and-Wait ARQ for flow control. The transmission delay and propagation delay (in millisecond) are 10 and 20. If channel bandwidth is 10Kbps then calculate throughtput (in Kbps)?

Q6 Consider two hosts connected directly through point-to-point link uses Sliding Window ARQ for flow control. The transmission delay and propagation delay (in millisecond) are 5 and 25. What should be optimal window size to achieve full utilization?

Q7 Consider two hosts connected directly through point-to-point link uses Go Back N ARQ for flow control. The transmission delay and propagation delay (in millisecond) are 2 and 255. What should be minimum number of bits required for sequence number field to achiev full utilization?

(A) 8 (B) 9
(C) 10 (D) 11

Q8 Consider two hosts connected directly through point-to-point link uses Selective Repeat ARQ for flow control. The transmission delay and propagation delay (in millisecond) are 4 and 200. What should be minimum number of bits required for sequence number field to achieve full utilization?

(A) 6 (B) 7

(C) 8 (D) 9

Android App | iOS App | PW Website

Answer Key

Q1	(D)	Q5	2
Q2	471	Q6	11
Q3	490	Q7	(B)
Q4	(B)	Q8	(C)

Hints & Solutions

Q1 Text Solution:

Packet size = 2000 bytes

D.T.R. = 1 Mbps

$t_x = \frac{\text{Packet size}}{\text{DTR}}$

$= \frac{2000 \text{ byte}}{1 \text{ mbps}}$

$= \frac{16\times10^3 \text{ bits}}{10^3\times10^3 \text{ bits / sec}}$ = 16 ms

Distance = 10 km

Signal speed = 20 ms/km

t_p = Distance × signal speed

= 10 km × 20 ms/km

= 200 ms

Q2 Text Solution:

File size = 24000 bytes

Packet size = 8000 bytes

$N = \frac{\text{File size}}{\text{Packet size}}$

N = 3

DTR = 1 Mbps

$t_x = \frac{8000 \text{ bytes}}{1 \text{ mbps}}$

$\frac{64\times10^3 \text{ bits}}{10^3\times10^3 \text{ bits / sec}}$

= 64 ms

Distance = 20 km

Signal speed = 2 × 10^5 m/sec

$t_p = \frac{\text{Distance}}{\text{Signal speed}} = \frac{20 \text{ km}}{2\times10^5 \text{m/ sec}}$

$\frac{2\times10^4 \text{ meter}}{2\times10^2\times10^3 \text{ meter / sec}}$ = 100 ms

Processing time of a packet = 15 ms

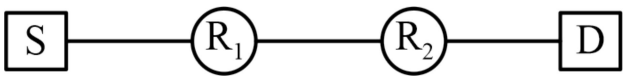
Total time

= (N × t_x + t_p) + processing time + (t_x + t_p)

= (3 × 64 + 100) + 15 + (64 + 100) ms

= 471 ms

Q3 Text Solution:



File size = 4000 bytes

Packet size = 1000 bytes

$N = \frac{4000 \text{ byte}}{1000 \text{ bits}}$

$= \frac{32\times10^3 \text{ bits}}{10^3 \text{ bits}}$ = 32

DTR = 100 kbps

$t_x = \frac{1000 \text{ bits}}{100 \text{ kbps}}$

$= \frac{10^3 \text{ bits}}{10^2\times10^3 \text{ bits / sec}}$

= 10 ms

Distance = 10 km

Signal speed = 5 ms/km

t_p = 10 km × 5 ms/km

= 50 ms

Total time = (N × t_x + t_p) + (t_x + t_p) + (t_x + t_p)

= (32 × 10 + 50) + (10 + 50) + (10 + 50) ms

= 490 ms

Q4 Text Solution:

Packet size = 1000 bytes

DTR = 1 Mbps

$T_x = \frac{1000 \text{ bytes}}{1 \text{ Mbps}}$

$= \frac{8\times10^3 \text{ bits}}{10^3\times10^3 \text{ bits / sec}}$ = 8 ms

Stop-and-wait ARQ

$\eta = \frac{t_x}{(t_x+2t_p)}$

$\left[\eta = \frac{1}{2}\right]$

$\frac{1}{2} = \frac{t_x}{(t_x+2t_p)}$

$(t_x + 2t_p) = 2t_x$

$t_p = \frac{t_x}{2} = 4 \text{ ms}$

Q5 Text Solution:

t_x = 10 ms

t_p = 20 ms

$\eta = \frac{t_x}{(t_x+2t_p)} = \frac{10 \text{ ms}}{(10+2\times20) \text{ ms}}$

$\eta = \frac{1}{5}$

Through put

= $\eta \times \text{DTR}$

= $\frac{1}{5} \times 10\,\text{kbps}$

= 2 kbps

Q6 Text Solution:

t_x = 5 ms

t_p = 25 ms

optimal window size = $\left\lceil \frac{(t_x+2t_p)}{t_x} \right\rceil$

= $\frac{(5+2\times25)\text{ ms}}{5\text{ ms}} = 11$

Q7 Text Solution:

t_x = 2 ms

t_p = 255 ms

optimal window size = $\left\lceil \frac{(t_x+2t_p)}{t_x} \right\rceil$

$\left\lceil \frac{(5+2*255)\text{ ms}}{2\text{ ms}} \right\rceil = 256$

N = 256

In go back NARQ

Total number of sequences = (n + 1) = 257

minimum number of bits required for sequence number field

= $\lceil \log_2 (\text{Total number of sequences}) \rceil$ bits

= $\lceil \log_2 (257) \rceil$ bits = 9 bits

Q8 Text Solution:

t_x = 4ms

t_p = 200 ms

optimal w.s = $\left\lceil \frac{(t_x+2t_p)}{t_x} \right\rceil = \left\lceil \frac{(4+2*200)}{4} \right\rceil$

$$N = \left\lceil \frac{404}{4} \right\rceil = 101$$

Selective repeate ARQ

Total number of sequences = 2N = (N + N) = 202

Minimum number of bits required for sequnces number field

= $\lceil \log_2 (\text{total number of sequnces}) \rceil$ bits

= $\lceil \log_2 (202) \rceil$ bits

= 8 bits